

CHEM 101 Topic 8: Chemical Quantities — Problem Set

Instructions: Complete the following problems using the analysis method practiced in your guided assignment. Show all your work, including conversion factors, molar mass calculations, and units at each step.

Section 1: Avogadro's Number and the Mole

Problem 1: A sample of pure Silver (Ag) contains 3.011×10^{23} atoms. How many moles of silver are present?

Problem 2: How many molecules are in 0.750 moles of Ammonia (NH₃)?

Problem 3: How many total atoms are in 2.00 moles of Sulfuric Acid (H₂SO₄)?

Section 2: Molar Mass and Mass-Mole Conversions

Problem 4: Calculate the molar mass of Potassium Nitrate (KNO₃).

Problem 5: A chemist weighs out 25.0 g of KNO₃. How many moles is this?

Problem 6: How many grams of Magnesium Chloride (MgCl₂) are required to obtain 0.125 moles of the compound?

Section 3: Molarity

Problem 7: Calculate the molarity of a solution containing 3.00 moles of solute dissolved in 1.50 L of solution.

Problem 8: What is the molarity of a solution prepared by dissolving 11.69 g of NaCl (molar mass 58.44 g/mol) in enough water to make 250 mL of solution?

Problem 9: An IV bag contains 2.00 L of 0.050 M Dextrose (C₆H₁₂O₆, molar mass 180.16 g/mol). How many grams of dextrose are in the bag?

Section 4: Empirical and Molecular Formulas

Problem 10: A compound has an empirical formula of NO and a molecular formula of N₂O₄. What is the multiplier?

Problem 11: A compound has an empirical formula of CH₂O (molar mass 30.03 g/mol) and a measured molar mass of 90.09 g/mol. What is the molecular formula?

Problem 12: A compound has an empirical formula of C₃H₅O (molar mass 57.07 g/mol) and a measured molar mass of 171.21 g/mol. Determine the molecular formula.